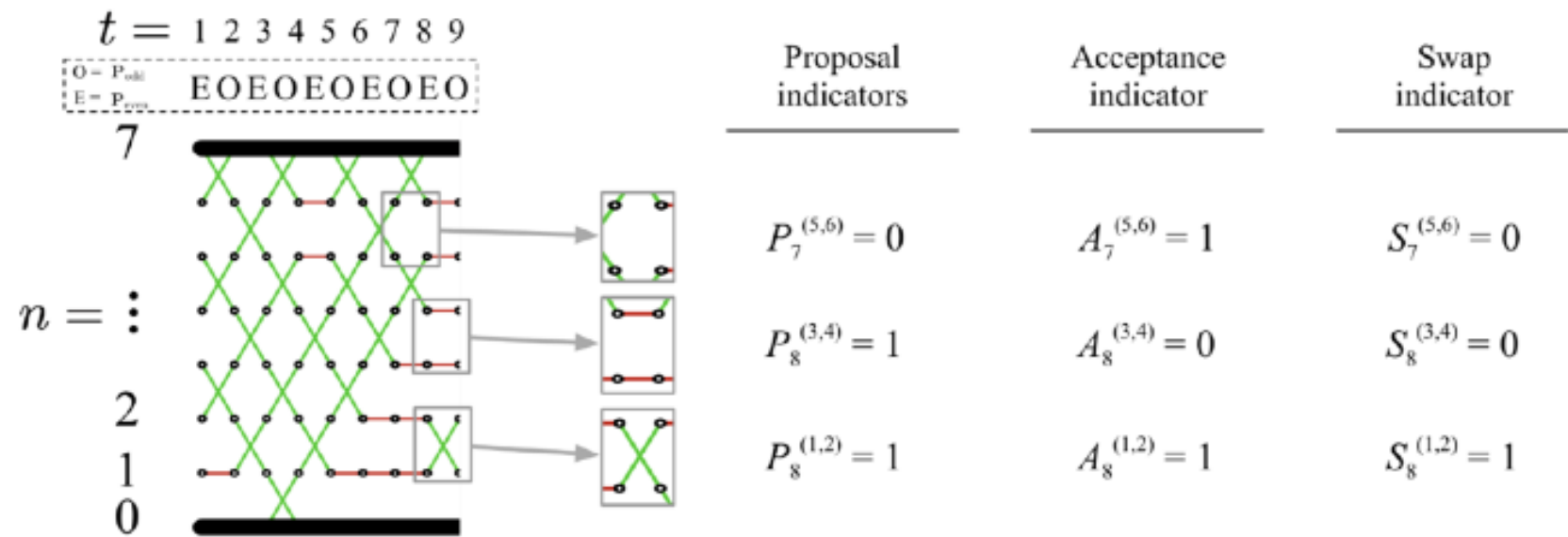




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► **Local exploration:** For each m update Y_t^m with the local move $K_{\beta_t^m}$ targeting $\pi_{\beta_t^m}$

► **Communication:** machine m proposes swaps of indexes I_{t-1}^m with machine storing $I_{t-1}^m + \epsilon_{t-1}^m$



DISTRIBUTED

$$Y_{t+1}^m \sim K_{\beta_t^m}(Y_t^m, dy)$$



anti-spiral

$$Y_0^m=X_0^m, \quad I_0^m=n, \quad \epsilon_0^m=(-1)^m$$

► If proposed swap is accepted:



Otherwise

$$I_{t+1}^m = I_t^m + \epsilon_t^m, \quad S_t^{(I_t^m, I_t^m + \epsilon_t^m)} = 1$$

$$\epsilon_t = 1, \quad P_{t+1}^{(I_t^m, I_{t+1}^m + 1)} = 1$$

► If the next scheduled proposal is an increase

E ↗

=

-

1

I^m

$t + 1$

=

=

I^m

t



Otherwise

DISTRIBUTED PT

► **Intialise:**

$$Y_0^m = X_0^m, \quad I_0^m = m, \quad \epsilon_0^m = (-1)^m$$

► **Local exploration:** For each m update Y_t^m with the local move $K_{\beta_t^m}$ targeting $\pi_{\beta_t^m}$

$$Y_{t+1}^m \sim K_{\beta_t^m}(Y_t^m, dy)$$

► **Communication:** machine m proposes swaps of indexes I_{t-1}^m with machine storing $I_{t-1}^m + \epsilon_{t-1}^m$

► If proposed swap is accepted:

$$I_{t+1}^m = I_t^m + \epsilon_t^m, \quad S_t^{(I_t^m, I_t^m + \epsilon_t^m)} = 1$$

► If the next scheduled proposal is an increase

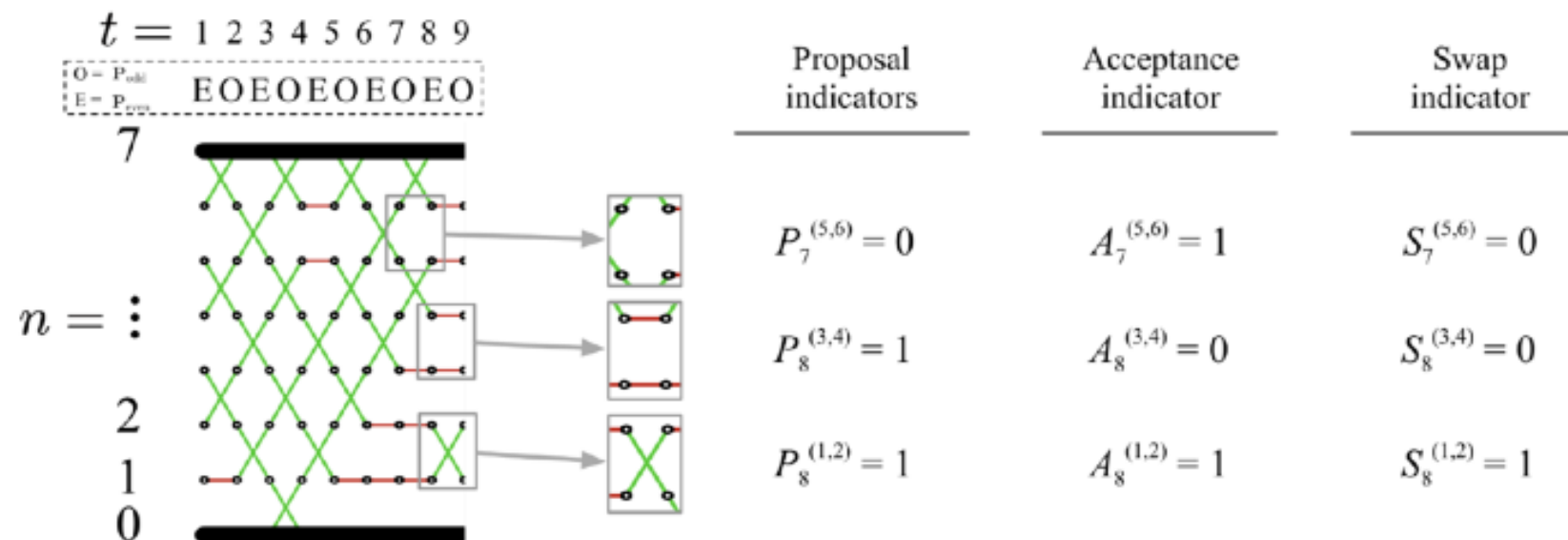
$$\epsilon_t = 1, \quad P_{t+1}^{(I_{t+1}^m, I_{t+1}^m + 1)} = 1$$

► Otherwise

$$I_{t+1}^m = I_t^m$$

► Otherwise

$$\epsilon_t = -1$$



OBJECTIVE OF PT